What is Regression

Regression is a statistical method used to analyze the relationship between a dependent variable and one or more independent variables.

It aims to identify the relationship between the variables and use that relationship to make predictions about the dependent variable.

It is a type of supervised learning technique in machine learning where the goal is to predict the value of a dependent variable based on the values of one or more independent variables.

Regression analysis is used in various fields such as finance, economics, social sciences, and engineering, among others, to model and predict the behavior of a dependent variable based on the values of independent variables.

There are different types of regression techniques, including simple linear regression, multiple linear regression, logistic regression, and polynomial regression.

Simple linear regression involves modeling the relationship between two variables using a straight line, while multiple linear regression involves modeling the relationship between one dependent variable and multiple independent variables.

Logistic regression is used when the dependent variable is categorical, and polynomial regression is used when the relationship between the variables is not linear but can be approximated using a polynomial function.

what is simple linear regression?

Simple linear regression is a type of regression analysis that models the relationship between two continuous variables, where one variable (the dependent variable) is assumed to be a linear function of the other variable (the independent variable). It is called "simple" because it involves only one independent variable.

In simple linear regression, a straight line is fitted to the data points to estimate the relationship between the two variables. The equation for a simple linear regression model can be written as:

$$y = \beta_0 + \beta_1 x + \varepsilon$$

where:

- y is the dependent variable
- x is the independent variable
- β_0 is the intercept term of y
- β_1 is the slope coefficient, which represents the change in y for a unit change in x
- ε is the error term or the random component of the model

The goal of simple linear regression is to estimate the values of β_0 and β_1 that best fit the data points and can be used to make predictions for new values of the independent variable.

The fit of the model can be assessed by measuring the amount of variation in the dependent variable that is explained by the independent variable, which is represented by the coefficient of determination (R-squared).

what is the best fit of line?

In regression analysis, the best-fit line is the line that most closely approximates the relationship between the dependent variable and the independent variable.

It is also known as the regression line, or the line of best fit.

The best-fit line is determined by minimizing the sum of the squared distances between the observed data points and the predicted values on the line.

This is also known as the sum of squared errors or residuals. The line that minimizes the sum of the squared errors is the line that best fits the data.

The best-fit line can be expressed mathematically as:

y = b0 + b1x

where

- y is the dependent variable,
- x is the independent variable,
- b0 is the intercept,
- b1 is the slope.

The slope represents the change in the dependent variable for each unit increase in the independent variable, while the intercept represents the value of the dependent variable when the independent variable is zero.

The best-fit line is important in regression analysis because it allows us to make predictions about the dependent variable based on the independent variable.

The closer the data points are to the line, the more accurate our predictions are likely to be.

what is intercept?

In the context of linear regression, the intercept, denoted by β0, is the value of the dependent variable when the independent variable is equal to zero.

It represents the point at which the regression line intersects the y-axis, and it is the value of the dependent variable when the independent variable has no effect on the value of the dependent variable.

In other words, the intercept is the value of y when x equals zero, assuming a linear relationship between the two variables. It can be interpreted as the baseline value of the dependent variable, which is not affected by changes in the independent variable.

In the simple linear regression model, the intercept and the slope coefficient are estimated from the data to represent the relationship between the two variables. The intercept is important because it helps to determine the starting point of the regression line and to interpret the relationship between the two variables.

The formula for the intercept in a simple linear regression model is:

$\beta_0 = \bar{y} - \beta_1 \bar{x}$

In linear regression, the intercept (β0) represents the expected value of the dependent variable (Y) when the independent variable (X) is equal to zero. It is also known as the y-intercept.

Example:

we have the number of data on the hours of study and exam scores of 10 students:

Hours Studied (x)	Exam Score (y)
3	70
5	85
2	60
8	90
6	85
1	50
7	95
4	75
9	95
5	80

We can perform a simple linear regression analysis to estimate the relationship between the hours of study and the exam scores.

The slope coefficient is found to be $\beta 1 = 8.33$

The intercept can be calculated using the formula:

$\beta 0 = \bar{y} - \beta 1\bar{x}$

where:

$\bar{y} = (70+85+60+90+85+50+95+75+95+80)/10 = 77$
$\bar{x} = (3+5+2+8+6+1+7+4+9+5)/10 = 5$

putting in the values, we get:

$\beta 0 = 77 - 8.33(5) = 38.35$

Therefore, the estimated intercept is 38.35, which means that if a student does not study at all (x=0), we would expect their exam score to be around 38.35, assuming a linear relationship between the two variables.

What is slope co-efficient?

In the context of linear regression, the slope coefficient, denoted by **β1**, represents the change in the dependent variable for a one-unit change in the independent variable.

It is the rate at which the dependent variable changes with respect to the independent variable, assuming a linear relationship between the two variables.

A positive slope indicates a positive relationship between the two variables, where an increase in the independent variable leads to an increase in the dependent variable, while a negative slope indicates a negative relationship, where an increase in the independent variable leads to a decrease in the dependent variable.

The slope coefficient is estimated from the data in the regression analysis and is used to determine the best-fit line that represents the relationship between the two variables.

To calculate the slope coefficient (β1) in a simple linear regression analysis, you can use the following formula:

$$\hat{Y} = b_0 + b_1 x$$

b_0 - the y-intercept, where the line crosses the y-axis.
b_1 - the slope, describes the line's direction and incline.

$$b_1 = \frac{SP_{xy}}{SS_x} = \frac{\Sigma(x_i - \bar{x})(y_i - \bar{y})}{\Sigma(x_i - \bar{x})^2}$$

$$b_0 = \bar{y} - b_1\bar{x}$$

The population correlation coefficient for X and Y is given by the formula:

$$\rho_{X,Y} = \mathrm{corr}(X,Y) = \frac{\mathrm{cov}(X,Y)}{\sigma_X \sigma_Y} = \frac{\mathrm{E}[(X-\mu_X)(Y-\mu_Y)]}{\sigma_X \sigma_Y}$$

Where,

ρ_{XY} = Population correlation coefficient between X and Y
μ_X = Mean of the variable X
μ_Y = Mean of the variable Y
σ_X = Standard deviation of X
σ_Y = Standard deviation of Y
E = Expected value operator
Cov = Covariance

The above formulas can also be written as:

$$\rho_{X,Y} = \frac{\mathrm{E}(XY) - \mathrm{E}(X)\,\mathrm{E}(Y)}{\sqrt{\mathrm{E}(X^2) - \mathrm{E}(X)^2} \cdot \sqrt{\mathrm{E}(Y^2) - \mathrm{E}(Y)^2}}$$

The sample correlation coefficient formula is:

$$r_{xy} = \frac{n\sum x_i y_i - \sum x_i \sum y_i}{\sqrt{n\sum x_i^2 - (\sum x_i)^2}\,\sqrt{n\sum y_i^2 - (\sum y_i)^2}}$$

Pearson Correlation Coefficient (r)

The Pearson correlation coefficient (*r*) is the most common way of measuring a linear correlation. It is a number between –1 and 1 that measures the strength and direction of the relationship between two variables.

Pearson correlation coefficient (*r*)	**Correlation type**	**Interpretation**	**Example**
Between 0 and 1	Positive correlation	When one variable changes, the other variable changes in the **same direction**.	Baby length & weight: The longer the baby, the heavier their weight.
0	No correlation	There is **no relationship** between the variables.	Car price & width of windshield wipers: The price of a car is not related to the width of its windshield wipers.
Between 0 and –1	Negative correlation	When one variable changes, the other variable changes in the **opposite direction**.	Elevation & air pressure: The higher the elevation, the lower the air pressure.

What is the Pearson correlation coefficient?

The Pearson correlation coefficient (*r*) is the most widely used correlation coefficient and is known by many names:

- Pearson's *r*
- Bivariate correlation
- Pearson product-moment correlation coefficient (PPMCC)
- The correlation coefficient

The Pearson correlation coefficient is a descriptive statistic, meaning that it summarizes the characteristics of a dataset. Specifically, it describes the strength and direction of the linear relationship between two quantitative variables.

Although interpretations of the relationship vary between disciplines

Pearson correlation coefficient (*r*) value	**Strength**	**Direction**
Greater than .5	Strong	Positive
Between .3 and .5	Moderate	Positive
Between 0 and .3	Weak	Positive
0	None	None
Between 0 and −.3	Weak	Negative
Between −.3 and −.5	Moderate	Negative
Less than −.5	Strong	Negative

Visualizing the Pearson correlation coefficient

Another way to think of the Pearson correlation coefficient (*r*) is as a measure of how close the observations are to a line of best fit.

The Pearson correlation coefficient also tells you whether the slope of the line of best fit is negative or positive. When the slope is negative, *r* is negative. When the slope is positive, *r* is positive.

When *r* is 1 or –1, all the points fall exactly on the line of best fit:

When *r* is greater than .5 or less than –.5, the points are close to the line of best fit:

When *r* is between 0 and .3 or between 0 and –.3, the points are far from the line of best fit:

When *r* is 0, a line of best fit is not helpful in describing the relationship between the variables:

When to use the Pearson correlation coefficient

The Pearson correlation coefficient (r) is one of several correlation coefficients that you need to choose between when you want to measure a correlation.

The Pearson correlation coefficient is a good choice when all of the following are true:

- **Both variables are quantitative:** You will need to use a different method if either of the variables is qualitative.
- **The variables are normally distributed:** You can create a histogram of each variable to verify whether the distributions are approximately normal. It's not a problem if the variables are a little non-normal.
- **The data have no outliers:** Outliers are observations that don't follow the same patterns as the rest of the data. A scatterplot is one way to check for outliers—look for points that are far away from the others.
- **The relationship is linear:** "Linear" means that the relationship between the two variables can be described reasonably well by a straight line. You can use a scatterplot to check whether the relationship between two variables is linear.

calculating the Pearson correlation coefficient (*r*):

$$r = \frac{n\sum xy - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$